SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

Course Code:	CSA06	**Course Title:**	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	**Assessment:**	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	**Total Marks:**	100 (2 Questions × 50 Marks)
CO1 Statement:	*Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)*		
Student Name:	T.Muthu Vaishnavi	**Reg. No.:**	192311215
Section / Batch:	Computer science	**Date:**	27.07.2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
	GRAND TOTAL:	**100 Marks**	

SECTION A – CONCEPT MAPPING

Code of Conduct

I T.Muthu Vaisnavi (192311215) certify that this submision is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: T.Muthu Vaisnavi

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

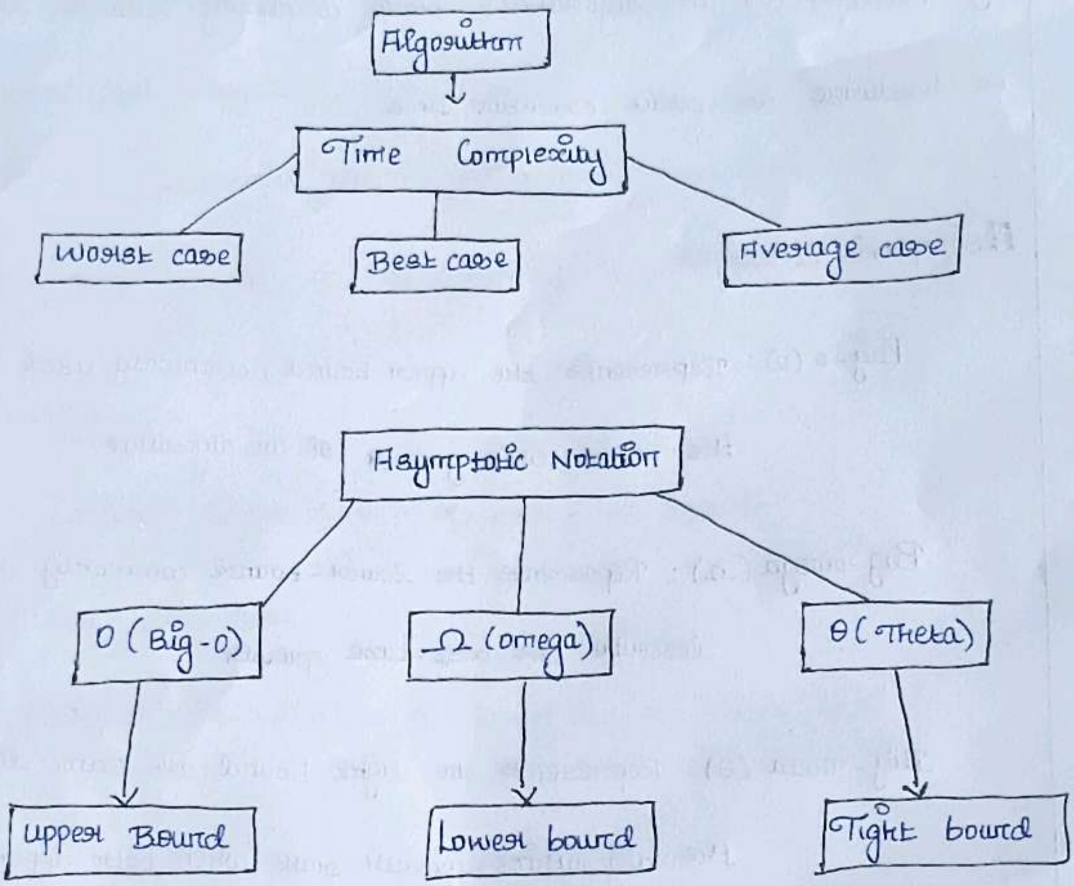
Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
					Overall Total (100)	

Faculty In-charge	Course Coordinator	Head of Department
Signature: ______________	Signature: ______________	Signature: ______________
Date: ________________	Date: ________________	Date: ________________

Assignment-2

Q. Concept Map : Algorithm -> Time complexity -> Asymptotic Notation

Concept Map



Explanation Of Relationships

Algorithm -> Time Complexity

An algorithm is a step by step procedure used to solve a problem. Time complexity measures how the running time of an algorithm grows as the input size (n) increases

Time Complexity -> Asymptotic Notation

Asymptotic notation is used to mathematically describe the growth rate of an algorithm's time complexity without depending on hardware or exact execution time

Asymptotic Notations

Big-o (O): Represents the upper bound, commonly used to describe the worst case growth of an algorithm

Big-omega (Ω): Represents the lower-bound commonly used to describe the best-case growth

Big-Theta (Θ): Represents the tight bound are same describing the algorithms growth rate when both upper and lower bounds are the same

Case Analysis

Best case analysis - Determines the minimum time required when the input is in the most favourable condition. It is often represented using Ω

Average - case analysis :

Determines the expected running time for typical or random inputs. It is often represented using θ

Worst case analysis

Determines the maximum time required for the most unfavorable input. It is commonly represented using O

Example

Consider linear search of array of size n

Case - Best case

Situation time - Element is found at the first position

Time - $\Omega(1)$

Case - Average case

Situation time - Element is found around in the middle of

Time - $\theta(n)$

Case - Worst case

Situation time - Element is the last position or absent

Time - $O(n)$

Conclusion

The concept map shows how an abstract algorithm is converted into a measurable efficiency metric through time complexity and asymptotic notation.

The best case, average case & worst case branches provide different perspectives on performance.